

Emotion Detection from Text · TF-IDF + Logistic Regression

Version 1.0 · October 2026

Model: EmoTFIDF-LR · 6-class, single-label · Data: `dair-ai/emotion` , config `split` (Hugging Face)

Official test set · n = 2,000

ACCURACY

88.7%

1,774 of 2,000 test texts correct

MACRO-F1

MAIN METRIC

0.845

all 6 emotions weighted equally

WEIGHTED-F1

0.890

macro P 0.823 · macro R 0.878

GAIN OVER BASELINES

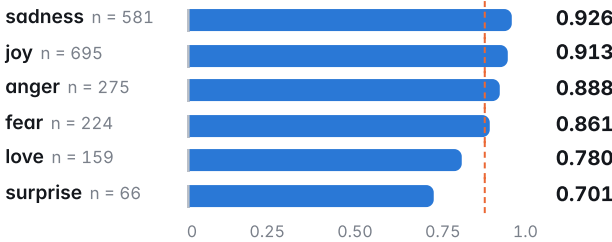
+0.41

macro-F1 vs Naive Bayes (0.435)

1 Model at a glance

Task	Main emotion of a short English sentence: sadness, joy, love, anger, fear, surprise (imbalanced, 9.4×).
Data	Official splits. Train 16,000 → 15,923 (77 duplicate, conflicting or test-leaking rows removed) · val 2,000 · test 2,000, untouched.
Pipeline	Lower-case, strip URLs/markup, join contractions, keep stop-words (so “not” survives) → TF-IDF unigrams, 7,226 words, fitted on train only.
Model	Logistic Regression, C = 3, balanced class weights. Best of 16 configs on validation macro-F1.
Use	✓ For teaching, research, aggregate trends with human review. ✗ Not for decisions about individuals (mental-health screening, hiring, moderation).

2 F1 score per emotion



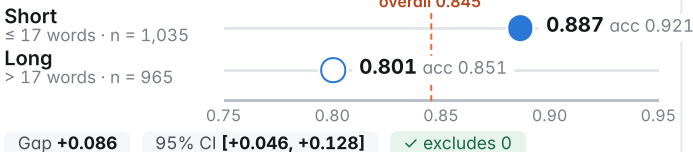
dashed line = macro-F1 0.845. Rare classes score lowest. Top error: **joy** → **love** (55 texts, 24% of errors).

3 Performance across two text subgroups / styles

macro-F1 on test set · 95% bootstrap CI, 1,000 resamples

A · Text length: short vs long

Split at the training-set median (17 words)

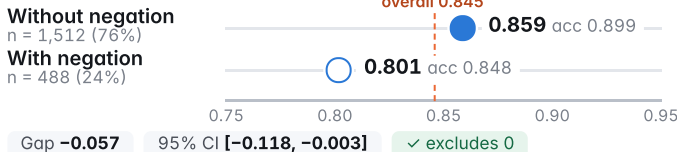


Finding: scores fall steadily as texts get longer. Long texts mix feelings, and bag-of-words adds up competing cues.

1–11 w **0.883** 12–17 **0.897** 18–25 **0.835** 26+ **0.772**

B · Negation style: with vs without

Contains ≥ 1 word from a fixed list: not, no, never, didnt, cant ...



Finding: mostly a length effect: 36% of long texts contain negation vs 13% of short ones. Controlling for length shrinks the gap:

within short **+0.017** within long **-0.033**

4 Explainability check: top features

centred Logistic Regression weights · longer bar = stronger push toward that emotion



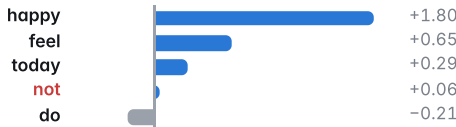
Reading: the model acts like a learned emotion lexicon: top features are feeling-adjectives, which is plausible. Some, like *superior* → *joy* or *dangerous* → *anger*, reflect this dataset's labelling habits. All bars share one 0–14 scale.

Word contributions for one sentence

“i do not feel happy today”

Predicted: **joy** (p = 0.86)

wrong: the sentence is negated



contribution = TF-IDF value × class weight (exact for a linear model)

58.8%

Occlusion test: deleting the single most influential word flips 58.8% of the 1,774 correct test predictions. **The model leans on one keyword.**

5 Limitations

observed in experiments unless marked “not measured”

▲ MAIN LIMITATION

The model does not understand negation

It treats text as a bag of words: “not” and “happy” are separate features with no word order. In a controlled probe, **6 of 6 negated sentences** were still classified as the emotion being negated, with high confidence.

i am not scared of the dark anymore → predicted **fear** (0.98)

i do not feel happy today → predicted **joy** (0.86)

- **Long, mixed-emotion texts** are harder: macro-F1 drops from 0.887 to 0.801 (Section 3A).
- **Rare and overlapping emotions:** surprise F1 0.701 (precision 0.614); joy and love are often confused.
- **Label noise:** 30 training sentences have conflicting labels; all 11 train-test duplicates disagree.
- **Narrow domain:** 98.8% of texts contain “feel”; may not transfer to reviews, news or code-mixed text. English only, emoji ignored.
- **Not measured:** sarcasm, and fairness across groups of people (the dataset has no demographic data).